

## **Problem Statement**

### **ELECTRICITY CONSUMPTION ANALYSIS USING ORACLE, TABLEAU AND FLASK**

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#### **1. Introduction**

Electricity is one of the most important resources for residential, commercial, industrial, and agricultural sectors. Every day, a huge amount of electricity consumption data is generated across different states and regions. Analysing this data manually is difficult, time-consuming, and prone to errors.

This project provides an interactive solution for analysing electricity consumption using Oracle Database, SQL, Tableau, and Flask. The system enables users to visualize electricity usage through dashboards, apply filters for detailed analysis, and access the visualizations through a Flask web application.

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#### **2. Problem Statement**

Organizations and government departments collect electricity consumption data from different states, regions, and sectors. However, analysing large datasets manually is inefficient. Users often face difficulties in identifying consumption trends, comparing regional usage, and generating meaningful insights.

There is a need for an automated system that stores electricity consumption data efficiently, generates interactive visualizations, and provides users with an easy-to-use web interface for accessing dashboards and reports.

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#### **3. Existing System**

The existing system mainly relies on spreadsheets and manual report generation.

##### **Drawbacks**

- Manual data analysis is time-consuming.
  - Difficult to identify trends and patterns.
  - Reports are not interactive.
  - Comparing multiple regions is difficult.
  - Decision-making takes more time.
  - Data visualization is limited.
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#### **4. Proposed System**

The proposed system stores electricity consumption data in an Oracle Database. SQL queries are used to retrieve and process the data. Tableau Desktop is used to create interactive dashboards and

stories. The dashboards are published to Tableau Public and embedded into a Flask web application, allowing users to access electricity consumption reports from a browser.

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## **5. Objectives**

The main objectives of the project are:

- To analyse electricity consumption data efficiently.
  - To store data using Oracle Database.
  - To retrieve information using SQL queries.
  - To create interactive Tableau dashboards.
  - To create Tableau stories for better data presentation.
  - To develop a Flask web application for visualization.
  - To provide users with easy access to reports and dashboards.
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## **6. Scope of the Project**

The project focuses on:

- Electricity consumption analysis.
- State-wise analysis.
- Region-wise analysis.
- Sector-wise analysis.
- Monthly consumption trend analysis.
- Interactive dashboard filtering.
- Web-based dashboard visualization.

Future enhancements may include real-time data integration, predictive analytics, and AI-based electricity consumption forecasting.

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## **7. Expected Outcome**

After implementing the proposed system:

- Users can analyse electricity consumption efficiently.
- Interactive dashboards improve data understanding.
- Tableau stories provide meaningful insights.
- Flask enables easy access through a web browser.
- Decision-making becomes faster and more effective.

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## 8. Technologies Used

| Technology      | Purpose              |
|-----------------|----------------------|
| Oracle Database | Data Storage         |
| SQL             | Data Retrieval       |
| Tableau Desktop | Dashboard Creation   |
| Tableau Public  | Dashboard Publishing |
| Python          | Backend Programming  |
| Flask           | Web Application      |
| HTML            | Web Page Structure   |
| CSS             | Web Page Styling     |
| GitHub          | Version Control      |

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## 9. Conclusion

The Electricity Consumption Analysis system provides a simple, interactive, and efficient solution for analyzing electricity usage across different states, regions, and sectors. By integrating Oracle Database, SQL, Tableau, and Flask, the project offers an effective platform for visualization and decision-making. The proposed solution reduces manual effort, improves data analysis, and enhances accessibility through a web-based interface.